

# Shopee Product Matching - Final Report (Multimodal)

Listings: 9,578 (train 5,764 / val 1,866 / test 1,948); split by product group, thresholds tuned on val, all numbers on test.  
Metric: Kaggle-style mean per-listing F1 over predicted match sets ("retrieval-F1").

## 1. Approach

Two-stage retrieve-and-rerank system. Stage 1: every listing retrieves its top-20 neighbours under each of 4 encoders (tfidf\_char, sbert, resnet50, clip\_img); the union forms the candidate pool (avoids  $O(n^2)$  classification; candidate-recall ceiling on test = 0.991). Stage 2: each candidate pair is described by 16 features (cosines of 3 text and 2 image encoders, perceptual-hash distance/equality/same-file, title length ratio, word Jaccard, number-token Jaccard and conflict flags) and scored by weighted score fusion or a gradient-boosted classifier. A threshold chosen on val turns scores into matches; optional graph post-processing (symmetry / connected components) uses the fact that matching is an equivalence relation.

**Final system (selected on val):** scorer = E3 GBM (all features); post-processing = symmetric; threshold = 0.3251.  
**Test retrieval-F1 = 0.9123** (precision 0.9339, recall 0.9277).

Baseline: equal-weight score fusion of the best single text cosine and the best single image cosine, cosine similarity, no learning, K=20 per encoder, threshold tuned on val. Baseline test F1 = 0.9050.

## 2. Experiments

Single-modality systems (same candidate pool):

| method         | modality | val_F1 | test_F1 | test_P | test_R |
|----------------|----------|--------|---------|--------|--------|
| cos_tfidf_word | text     | 0.8481 | 0.8511  | 0.9000 | 0.8649 |
| cos_tfidf_char | text     | 0.8599 | 0.8534  | 0.8892 | 0.8790 |
| cos_sbert      | text     | 0.6771 | 0.6997  | 0.8397 | 0.6861 |
| cos_resnet50   | image    | 0.7027 | 0.7076  | 0.9485 | 0.6409 |
| cos_clip_img   | image    | 0.6928 | 0.7044  | 0.9434 | 0.6411 |

### E1 - Weighted score fusion

**Hypothesis.** Fusing text+image beats the best single modality and the optimal weight differs from 0.5.

**Experiment.** Sweep  $w$  in  $[0,1]$  on val for  $s=w*\text{text}+(1-w)*\text{image}$ ; choose threshold on val.

**Result.**  $w^*=0.5$ ; test F1 0.9050 vs baseline 0.9050 vs best single 0.8534 -> **NOT SUPPORTED (no gain)**

**Analysis.** The weight curve shows how much each modality contributes; a curve that peaks strictly between 0 and 1 means the modalities carry complementary information.

**Conclusion.** Hypothesis not supported (no gain).

### E1b - Ensembling within each modality

**Hypothesis.** Mean of several encoders per modality beats the single best encoder per modality.

**Experiment.**  $s = w*\text{mean}(\text{text cosines})+(1-w)*\text{mean}(\text{image cosines})$ ;  $w$ , thr tuned on val.

**Result.** test F1 0.9004 vs E1 0.9050 -> **REJECTED (worse)**

**Analysis.** Lexical (TF-IDF) and semantic (SBERT) text encoders, and CNN vs CLIP image encoders, err differently; their average is smoother.

**Conclusion.** Hypothesis rejected (worse).

### E2 - Learned (GBM) fusion on cosine features

**Hypothesis.** A non-linear learned combiner beats the best weighted-sum fusion.

**Experiment.** HistGradientBoosting on the same 5 cosines, trained on train-split candidate pairs.

**Result.** test F1 0.9063 vs linear 0.9050 -> **NOT SUPPORTED (no gain)**

**Analysis.** A GBM can learn interactions (e.g. both-modalities-agree vs one-modality-only) and per-feature non-linear calibration that a single weight cannot.

**Conclusion.** Hypothesis not supported (no gain).

### E3 - Side-information features (pHash, numbers, lengths)

**Hypothesis.** Variant/duplicate features reduce false positives and improve F1 over cosines only.

**Experiment.** GBM with 16 features (cosines + pHash + meta) vs GBM with 5 cosines.

**Result.** test F1 0.9105 vs 0.9063 -> **SUPPORTED**

**Analysis.** Number-token conflicts separate size/model variants; pHash==0 identifies re-used pictures; word-overlap and length ratios flag spammy or truncated titles.

**Conclusion.** Hypothesis supported.

## E4 - Graph post-processing

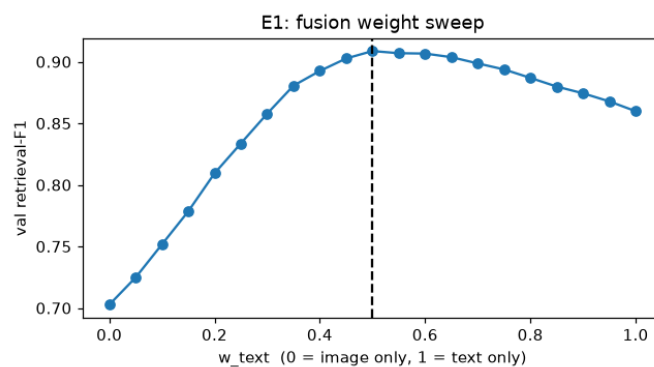
**Hypothesis.** Symmetrising / transitive closure raises recall more than it hurts precision.

**Experiment.** none vs symmetric vs connected components on E3 scores; thr tuned on val per variant.

**Result.** [{'post-processing': 'none', 'thr(val)': 0.2784, 'val\_F1': 0.9106, 'test\_F1': 0.9105, 'test\_P': 0.9275, 'test\_R': 0.9314}, {'post-processing': 'symmetric', 'thr(val)': 0.3251, 'val\_F1': 0.9112, 'test\_F1': 0.9123, 'test\_P': 0.9339, 'test\_R': 0.9277}, {'post-processing': 'components', 'thr(val)': 0.4748, 'val\_F1': 0.9055, 'test\_F1': 0.8861, 'test\_P': 0.8829, 'test\_R': 0.95}] -> **NOT SUPPORTED (no gain)**

**Analysis.** Closure helps products with many listings (recall) but a single false edge can merge two products (precision); the max component size guard limits the damage.

**Conclusion.** Best on val: symmetric; hypothesis not supported (no gain).



| method                                       | val_F1 | test_F1 | test_P | test_R |
|----------------------------------------------|--------|---------|--------|--------|
| BASELINE: 0.5*text + 0.5*image               | 0.9087 | 0.9050  | 0.9324 | 0.9221 |
| E1 weighted fusion (w_text=0.5)              | 0.9087 | 0.9050  | 0.9324 | 0.9221 |
| E1b ensemble-in-modality fusion (w_text=0.5) | 0.9082 | 0.9004  | 0.9124 | 0.9341 |
| E2 GBM on 5 cosine features                  | 0.9100 | 0.9063  | 0.9229 | 0.9303 |
| E3 GBM on all 16 features                    | 0.9106 | 0.9105  | 0.9275 | 0.9314 |
| FINAL (E3 GBM (all features) + symmetric)    | 0.9112 | 0.9123  | 0.9339 | 0.9277 |

## 3. Analysis

The best single encoder is cos\_tfidf\_char (0.8534); the final multimodal system reaches 0.9123 (+0.0589 absolute). For reference, the best text-only system of Part B reached 0.8507 and the best image-only system of Part C 0.7419 (full-split top-50 retrieval). Experiments marked SUPPORTED above are those where the hypothesis held on the test split; for REJECTED/NOT SUPPORTED ones the extra complexity did not pay off, which is itself a finding: components are only kept in the final system if they improved val F1. Text and image encoders err on different pairs (see error categories), which is why fusion helps; side-information features mostly act on false positives (variants with conflicting numbers, re-used pictures).

## 4. Error analysis

Stage-1 misses: 283 of 9,854 true directed pairs (2.9%) never enter the candidate pool and cannot be recovered.

| error category                                                               | count | why it fails                                               | possible fix                                                                                                |
|------------------------------------------------------------------------------|-------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| FN: neither modality matches (hard)                                          | 565   | both text and image differ - maybe even label noise        | more modalities/attributes, cluster-level reasoning, inspect label noise                                    |
| FP: text matches, images differ                                              | 398   | generic or copied titles; seller photos differ             | down-weight generic title words, require image agreement when titles are generic                            |
| FN: titles match, images differ (different photos of same product)           | 311   | viewpoint, packaging vs item, lifestyle photo              | image fine-tuning with multi-view positives, rely more on text when the image is unreliable                 |
| FP: text AND image agree (near-identical variant / similar product)          | 132   | same product family, different variant (colour/size/model) | variant-aware features: colour histogram, size/number extraction, fine-tuned embeddings with hard negatives |
| FP: fused model over-trusted weak cues                                       | 131   | many individually weak signals added up                    | calibrate scorer, raise threshold, add more negatives in training                                           |
| FN: image matches, titles differ (noisy / disjoint titles)                   | 122   | other language, keyword spam, very short title             | multilingual encoder, title cleaning, translation, trust image more for short titles                        |
| FP: image matches, titles differ (visually similar product)                  | 74    | stock photo / similar item                                 | require some text support; instance-level image embeddings                                                  |
| FN: both modalities match but fused model rejected (threshold / calibration) | 28    | fused score borderline                                     | threshold per group size / local adaptive thresholds                                                        |

#### Error rates for interpretable slices:

| slice                                                          | type | n_pairs | error_rate |
|----------------------------------------------------------------|------|---------|------------|
| all true matches                                               | FN   | 9571    | 0.1072     |
| true matches with a very short title ( $\leq 2$ words)         | FN   | 113     | 0.2035     |
| true matches with very different images (pHash Ham $\geq 24$ ) | FN   | 5721    | 0.1689     |
| true matches with near-identical images (pHash Ham $\leq 4$ )  | FN   | 2357    | 0.0000     |
| true matches whose number tokens conflict                      | FN   | 2331    | 0.2462     |
| all true non-matches                                           | FP   | 106474  | 0.0069     |
| non-matches whose number tokens conflict (variants)            | FP   | 39394   | 0.0065     |
| non-matches with high title word overlap (Jaccard $\geq 0.6$ ) | FP   | 32      | 1.0000     |

FP examples (predicted match; truth is the opposite)

text matches, images differ  
 score=0.97 thr=0.33  
 text=0.87 img=0.45  
 TRUE: different product



DT Shop.Medan Pintar Facil Paint Roller / Kuas Cat Tembok Rumah



text matches, images differ  
 score=0.97 thr=0.33  
 text=0.87 img=0.45  
 TRUE: different product



Pintar Facil Paint Roller / Kuas Cat Tembok Rumah



text AND image agree (near-identical)  
 score=0.99 thr=0.33  
 text=0.91 img=0.81  
 TRUE: diff



Tripod Portable Tefeng TF-3110 Panjang 1 Meter Ringan / Kaki Tripod TEFENG TF-3110 / Standing Holder



erent product



Tripod Portable Tefeng TF-3110 Panjang 1 Meter Ringan / Kaki Tripod TEFENG TF-3110 / Standing Holder



Tripod Portable Tefeng TF-3110 Panjang 1 Meter Ringan

text AND image agree (near-identical v score =0.99 thr=0.33 text=0.91 img=0.81 TRUE: different product



Mi Boncabe Level 15 / Mi Boncabe M urah / Mi Boncabe Grosir (READY STOCK) SIAP KIRIM READY STOCK !!



MIE BON CABA LEVEL 15

fused model over-trusted weak cues score=0.80 thr=0.33 text=0.36 img=0.61 TRUE: different product



MIE BON CABA LEVEL 15



Mi Boncabe Level 15 / Mi Boncabe M urah / Mi Boncabe Grosir (READY STOCK) SIAP KIRIM READY STOCK !!



fused model over-trust  
ed weak cues  
score=0.8  
0 thr=0.33  
text=0.36 i  
mg=0.61  
TRUE: differen  
t product



NIVEA Extra White Radiant & Smooth  
Lotion 200 ml



Nivea daily protection sun lotion  
light feeling 100ml



image matches, titles  
differ (visually  
score  
=0.83 thr=0.33  
text=0.  
38 img=0.85  
TRUE: diff  
erent product

Nivea daily protection sun lotion  
light feeling 100ml



NIVEA Extra White Radiant & Smooth  
Lotion 200 ml



image matches, titles  
differ (visually  
score  
=0.83 thr=0.33  
text=0.  
38 img=0.85  
TRUE: diff  
erent product

FN examples (predicted no match; truth is the opposite)

neither modality match  
es (hard)  
score=0.00 t  
hr=0.33  
text=0.22 img=  
0.29  
TRUE: same produc  
t



neither modality match  
es (hard)  
score=0.00 t  
hr=0.33  
text=0.22 img=  
0.29  
TRUE: same produc  
t



titles match, images d  
iffer (different  
score  
=0.00 thr=0.33  
text=0.  
44 img=0.29  
TRUE: same



TRUE: same product



Headset Stereo U19 Macaron Handsfree Extra Bass Earphone U 19



U19 Headset Macaron Mate Color HIFI Over Ear Suara Musik Earphones Android & iOS Universal Dengan Kabel...

titles match, images differ (different score)  
=0.00 thr=0.33  
text=0.44  
img=0.29  
TRUE: same product



image matches, titles differ (noisy / score)  
=0.00 thr=0.33  
text=0.04  
img=0.73  
TRUE: same product

Women drops BEST SELLER!



STIMULANT



STIMULANT



Women drops BEST SELLER!



image matches, titles  
differ (noisy /  
score  
=0.00 thr=0.33  
text=0,  
04 img=0.73  
TRUE: same  
product



Casing Soft Case Koper Motif Chrys  
anthemum Putih untuk OPPO A53 2020  
Reno 4 A5S A5 2020 A3S A92 A31 A1...

Casing Soft Case TPU oppo A92 A52  
Realme C15 C11 A12 A31(2020) Reno2  
F A9 2020 A5 2020 a71 a3s a37 F7 A...

both modalities match  
but fused model  
score  
=0.01 thr=0.33  
text=0,  
71 img=0.73  
TRUE: same  
product



Casing Soft Case TPU oppo A92 A52  
Realme C15 C11 A12 A31(2020) Reno2  
F A9 2020 A5 2020 a71 a3s a37 F7 A...

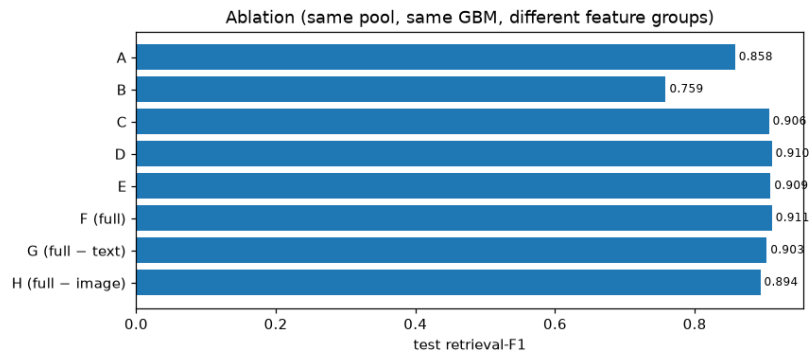
Casing Soft Case Koper Motif Chrys  
anthemum Putih untuk OPPO A53 2020  
Reno 4 A5S A5 2020 A3S A92 A31 A1...

both modalities match  
but fused model  
score  
=0.01 thr=0.33  
text=0,  
71 img=0.73  
TRUE: same  
product



## 5. Ablation study

| Configuration    | Text | Image | pHash | Meta<br>(nums/len) | val_F1 | Score<br>(test F1) | Precision | Recall | pair AP |
|------------------|------|-------|-------|--------------------|--------|--------------------|-----------|--------|---------|
| A                | yes  | no    | no    | no                 | 0.8614 | 0.8578             | 0.9030    | 0.8721 | 0.9156  |
| B                | no   | yes   | no    | no                 | 0.7402 | 0.7585             | 0.9332    | 0.7158 | 0.7345  |
| C                | yes  | yes   | no    | no                 | 0.9100 | 0.9063             | 0.9229    | 0.9303 | 0.9594  |
| D                | yes  | yes   | yes   | no                 | 0.9077 | 0.9104             | 0.9337    | 0.9258 | 0.9596  |
| E                | yes  | yes   | no    | yes                | 0.9119 | 0.9087             | 0.9225    | 0.9333 | 0.9597  |
| F (full)         | yes  | yes   | yes   | yes                | 0.9106 | 0.9105             | 0.9275    | 0.9314 | 0.9601  |
| G (full - text)  | no   | yes   | yes   | yes                | 0.8918 | 0.9032             | 0.9326    | 0.9148 | 0.9482  |
| H (full - image) | yes  | no    | yes   | yes                | 0.8890 | 0.8940             | 0.9301    | 0.9017 | 0.9480  |



## 6. Limitations

Frozen pretrained encoders (no fine-tuning for product identity); fixed top-K candidate pool bounds recall; a single global threshold regardless of group size; TF-IDF is fitted on the evaluated corpus (unsupervised, transductive); possible label noise; results from one random group split without confidence intervals; no explicit colour/size/brand attribute extraction.

## 7. Future improvements

Metric-learning fine-tuning (ArcFace/contrastive with hard-negative mining) of image and text encoders; joint multimodal embeddings; attribute extraction with conflict rules; HNSW index for scale; database-side augmentation / query expansion; adaptive thresholds per neighbourhood; cross-validated group splits with confidence intervals; community detection instead of plain connected components.

## 8. External resources

scikit-learn (TF-IDF, HistGradientBoosting); sentence-transformers paraphrase-multilingual-MiniLM-L12-v2 (Reimers & Gurevych, 2019); OpenAI CLIP ViT-B/32 via HuggingFace transformers (Radford et al., 2021); torchvision ResNet-50 / EfficientNet-B0 (ImageNet); Kaggle 'Shopee - Price Match Guarantee' dataset and its F1 metric; perceptual hashes provided in the dataset.